1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L26 –**
Review-Based Code Testing

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Recap

5 **Testing Methods**

- TWO broad groups of approaches
  - Review-based
  - Execution-based

6 **Review-Based Testing**

- TWO types
  - Code walkthrough (similar to cognitive walkthrough)
  - Code inspection (similar to heuristic evaluation)

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Code Walkthrough

8 **Basics**

- An informal code analysis technique
- Main objective - discover algorithmic and logical errors in the code

9 **Steps**

- Few members of development team are given code to read and understand

10 **Steps**

- Each member
  - Selects some test cases
  - Simulates execution by hand (i.e. trace execution through each statement and function execution)

11 **Steps**

- Members note down their findings and discuss in a walkthrough meeting in presence of coder(s)

12 **(Some) Guidelines**

- Team performing walkthrough should not be either too big or too small (ideally, 3-5 members)

13 **(Some) Guidelines**

- Use 'representative' test cases – similar to 'representative' use cases in cognitive walkthrough

14 **(Some) Guidelines**

- Discussion should focus on discovery of errors and not how to fix errors

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Code Inspection

16 **Basics**

- Aim is to discover some common types of errors caused due to oversight and improper programming (e.g., uninitialized variables)
- Also checks adherence to coding standards

17 **Basics**

- Typically companies collect statistics regarding different types of errors commonly committed by their engineers and identify type of errors most frequently committed
  - Such a list can be used during code inspection
- Similar to heuristic evaluation (evaluation with a checklist)

18 **Some Common Errors**

- Use of uninitialized variables
- Jumps into loops
- Nonterminating loops
- Incompatible assignments

19 **Some Common Errors**

- Improper storage allocation and deallocation
- Use of incorrect logical operators
- Incorrect precedence among operators
- Improper modification of loop variables

20 **Reference**

- Rajib Mall (2018). Fundamentals of Software Engineering, 5th ed, PHI Learning Pvt Ltd. Chapter 10
- Roger S Pressman (2020). Software Engineering: A Practitioner's Approach, 9th ed, McGraw-Hill Education, New York, Chapter 19-21